11.1 Evaluating Expressions

Lesson Introduction

Benchmark	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.	Learning Target	• I can apply order of operations to an algebraic expression.

	Building On	Working Toward
Benchmark Coherence	MA.5.AR.2 Demonstrate an understanding of equality, the order of operations, and equivalent numerical expressions.	MA.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

Essential Questions	• How do you evaluate an expression by applying the order of operations?
Lesson Narrative	In this first lesson of the unit, students extend their understanding of order of operations to evaluate numeric expressions from Grade 5 mathematics. Students connect numeric expressions with and without grouping symbols to mathematical models, deepening their understanding of these symbols. The term "variable" is introduced as students learn to evaluate algebraic expressions by substituting values in for the variables and then evaluating the numeric expressions.

Component Summary	11.1.1 Warm-Up (~5 min.)	Evaluate a numeric expression to resolve a disagreement *(SE p. 239)*	11.1.4 Guided Practice (5-10 min)	Evaluate algebraic expressions using order of operations *(SE p. 243)*
	11.1.2 Collaboration (10-15 min.)	Connect numerical expressions to mathematical models *(SE p. 240)*	11.1.5 Your Turn! (5-10 min)	Practice evaluating algebraic expressions using order of operations *(SE p. 243)*
	11.1.3 Guided Instruction (15-25 min.)	Extend evaluating numeric expressions using the order of operations to algebraic expressions *(SE p. 241)*	11.1.6 Wrap-Up (~5 min)	Demonstrate understanding of evaluating algebraic expressions using order of operations *(SE p. 244)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Variable • Evaluate • Value • Parentheses Additional Terms: N/A	• (Optional) Mini whiteboards • (Optional) Colored chips	• Consider providing small bags of math chips for students to utilize in small groups to represent and model expressions. These can be play chips, poker chips, or laminated cutouts.

Teacher Tips

Common Misconceptions

- Students may substitute the variable as a digit instead of multiplying it by the coefficient. For example, in $5x$, where $x = 4$, students may write 54 instead of $5(4)$.
- Students may prioritize left to right operations in lieu of parenthesis. For example, during the second expression in 11.1.2 Collaboration, $3 \times (5 + 4)$, students may multiply 3×5 first.

Things to Consider

- This lesson has several key glossary terms necessary for comprehension. Consider having students create interactive notebooks or a Frayer Model for vocabulary.
- Consider modeling examples and allowing students to demonstrate their mastery with chips as a way to make sense of expressions.
- There are many misconceptions about the order of operations. The teaching of the mnemonic device of PEMDAS leads students to perceive that the order of PEMDAS is rigid, where it is not. For example, in the expression 30 – 4(3 + 7) + 12 ÷ 3, there are options as to where to begin. Students actually have a choice and may first simplify the 3 + 7 in the parentheses, distribute the 4 to the 3 and to the 7, or perform 12 ÷ 3 before doing any other computation—all without affecting an accurate outcome. For this reason, the lessons do not empathize this mnemonic device. Encourage your students to consider different entry points and discuss them

Literacy and Language Routines

Poll the Class

11.1.2 Collaboration provides an opportunity to engage students in a Poll the Class routine to analyze two responses. This is an opportunity to engage all students and build interest and investment in this first lesson of the unit. Provide mini whiteboards for students or have them represent their responses by showing a number with their fingers.

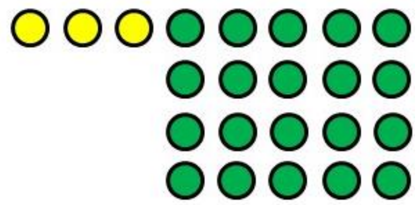
Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.4.1: Discussion and Reflection

11.1.3 Guided Instruction opens by presenting students with three hypothetical evaluations of an expression and calls on them to find which evaluation is correct. Clarifications for this standard include recognizing an error and suggesting how to fix it. Encourage students to not only find the error but formalize how to prevent it from happening in the future.

Differentiate Instruction

It may be helpful to provide students with colored chips to model the expressions in 11.1.2 Collaboration. This will provide a kinesthetic and visual entry point. Reading the expressions with their operations will help English Language Learners and visually impaired students by providing an auditory entry point.



Allow student choice in 11.1.5 Your Turn!. Students who may struggle to evaluate expressions can choose one to two to focus on to practice before moving on to the 11.1.6 Wrap-Up.

Lesson Notes:

11.1.1 Warm-Up: Evaluating Numeric Expressions

Component Overview: *Students examine two different values to determine which one correctly represents the expression, then work to find the value of another expression independently.*

11.1.1 Warm-Up: Evaluating Numeric Expressions

This math expression is dividing the internet.

What is the value of $8 \div 2(2+2)$?

Poll the Class
Engage students with a Poll the Class routine to determine whose answer they agreed with. Consider having students write their responses on a whiteboard and holding it up or show either one or two (using their hand) to indicate who they agree with. Encourage students to discuss how they know who is correct and the error the other person made.

1. Do you agree with Alex or Ms. B? Explain or show why using order of operations.
   Ms. B

$$8 \div 2(2+2)$$
$$8 \div 2(4)$$
$$4(4)$$
$$16$$

2. Find the value of the expression. Show your work.

$$4 - 2 \times 6 \div 3 + 7$$
$$4 + (-2) \times 6 \div 3 + 7$$
$$4 + (-12) \div 3 + 7$$
$$4 + (-4) + 7$$
$$4 - 4 + 7$$
$$0 + 7$$
$$7$$

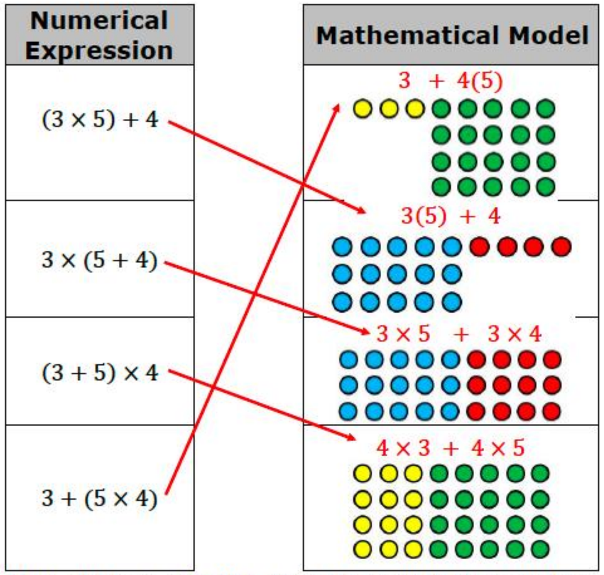
Students may not have social media, but they could recreate the activity for homework by polling their friends and family.

11.1.2 Collaboration: Modeling the Meaning of Expressions

Component Overview: *Students work with a partner to match numerical expressions to corresponding mathematical models. Then students work independently to determine the matching mathematical models for the expressions without the grouping symbols. Engage the class in a discussion to clarify how the mathematical models change when the symbols are removed.*

11.1.2 Collaboration: Modeling the Meaning of Expressions

1. Work with your partner to determine which mathematical model matches each numerical expression. Use arrows to indicate matches. Be prepared to explain your thinking.



2. How did you use the grouping symbols to help you determine which model matched each expression?
Answers will vary.

$(3 \times 5) + 4$
3 *groups of* $5 + 4$

$3 \times (5 + 4)$
$3 \times (9)$
3 *groups of* 9 *or* $3 \times 5 + 3 \times 4$

$(3 + 5) \times 4$
8×4
4 *groups of* 8
or $4 \times 3 + 4 \times 5$

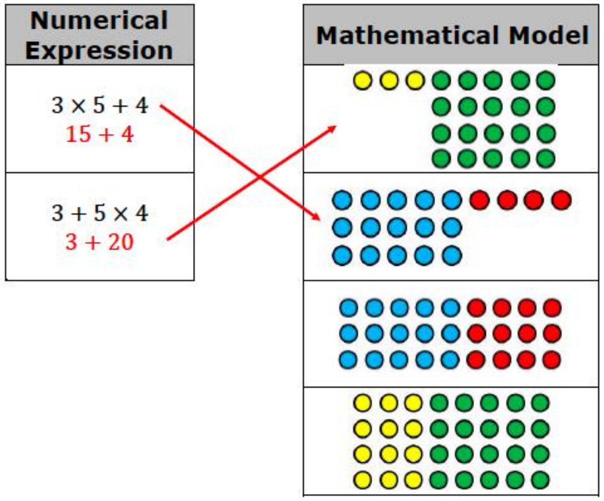
$3 + (5 \times 4)$
5 *groups of* 4 *plus* 3

Turn and Talk
Consider engaging students in a Turn and Talk routine before students write their responses for question 2. Have them turn and tell their partner how they used grouping symbols to determine each match. This will give students an opportunity to verbalize their responses before written responses, engaging auditory learners.

- *How can we distinguish between multiplication and addition in the models shown?*
- *What are grouping symbols? How are they used? Where do they fall in the order of operations?*
- *How can you show this using a model?*

11.1.2 Collaboration: Modeling the Meaning of Expressions (continued)

3. If the grouping symbols were removed, which model matches each expression? Use arrows to indicate matches.



Consider providing different colored chips for students to model each expression with before matching. This will provide visual and kinesthetic entry points.

What do you notice about the visual model when the grouping symbols are removed?

Engage the class in a discussion to determine how removing the symbols did not change the visual models of each expression because of the order of operations and focus on how this will impact the evaluation of each expression. Encourage students to refer back to question 1 if they are not sure how to respond or display both questions side by side so students can visually engage with the material.

11.1.3 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Component Overview: *Activate students' prior knowledge both from previous units and from this lesson in 11.1.1. Warm-Up and 11.1.2 Collaboration. Position students as sense makers in this error analysis scenario in order to elicit increasingly precise and mathematical responses on evaluating expressions.*

11.1.3 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations. Grouping symbols such as parentheses are used to indicate which operations should be performed first. The following order is used.

Step 1: Perform operations within grouping symbols.
Step 2: Multiply and divide in order from left to right (→).
Step 3: Add and subtract in order from left to right (→).

1. Three students completed the warm-up. The steps in each student's process are shown.

Leila's Work	Fabian's Work	Jaylah's Work
$4 - 2 \times 6 \div 3 + 7$	$4 - 2 \times 6 \div 3 + 7$	$4 - 2 \times 6 \div 3 + 7$
$2 \times 6 \div 3 + 7$	$4 - 12 \div 3 + 7$	$4 - 12 \div 3 + 7$
$12 \div 3 + 7$	$4 - 4 + 7$	$4 - 4 + 7$
$4 + 7$	$4 - 11$	$0 + 7$
11	-7	7

a. Complete the statements about which student correctly applied the given steps of the order of operations.

- Leila
- Fabian
- Jaylah

evaluated the numerical expression correctly.

She first performed all
- multiplication and division
- addition and subtraction

in order from left to right and then performed all
- multiplication and division
- addition and subtraction

in order from left to right.

Consider creating an anchor chart, color-coded and with visuals, as students review the order of operations they learned in 5th grade. This will provide a reference for visual learners and reinforce what was learned in earlier grades.

Error Analysis
Engage students in an error analysis of each student's work from question 1. Consider having students partner up to analyze the student work and determine the errors before participating in a whole class discussion to address the misconceptions displayed in each response.

Challenge students to not only name the errors each student made but to also write an explanation for how to fix and prevent them from making the same mistakes in the future.

11.1.3 Guided Instruction: Applying Order of Operations to Algebraic Expressions (continued)

b. With your partner, discuss the work of each student, Leila, Fabian, and Jaylah. Two of these students made a mistake somewhere. Circle the mistakes that you and your partner found.
Leila went from left to right.
Fabian did addition before subtraction.

2. Omari evaluated Expression A and Expression B. His work is shown.

Expression A	Expression B
$12 + 24 \div 6 - 2 \times 2$	$12 + 24 \div (6 - 2 \times 2)$
$12 + 4 - 2 \times 2$	$12 + 24 \div (6 - 4)$
$12 + 4 - 4$	$12 + 24 \div (2)$
$16 - 4$	$12 + 12$
12	24

a. With your partner, discuss the two original expressions that Omari evaluated and his corresponding answers.

b. Omari's partner gave him a 👍 on both of his answers. Why did Omari arrive at two different values when evaluating these expressions?
Omari had two different answers because Expression B contains parenthesis that are not in Expression A. Following order of operations Expression B is completed differently than Expression A.

Like numeric expressions, algebraic expressions can be evaluated using order of operations when given a specific value for the ***variable***(s).

In an algebraic expression, a ***variable*** is a letter that represents a number. It is called a **vari**able because the value used to replace the letter can ***vary***.

Turn and Talk
Before answering question 2, have students participate in a Turn and Talk routine to formalize their answers before circling the mistakes. Encourage students to notice the differences between the two expressions and consider how/why that might impact the value of the expressions.

- *What is the same about the expressions? What is different?*
- *How do parentheses impact the order of operations? How do they impact the value of the expression?*

Consider having students create a foldable, Frayer Model, or other type of interactive to allow students to synthesize the vocabulary outlined in this section. Examples, nonexamples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

11.1.3 Guided Instruction: Applying Order of Operations to Algebraic Expressions (continued)

3. When evaluating algebraic expressions, first substitute the given *number* for the variable(s). Then, apply the given steps for order of operations.

In mathematics, **evaluate** is an instruction that indicates the answer is a **value**.

4. Evaluate $\frac{n}{6} + 2$ when $n = 12$.

$$\frac{12}{6} + 2 = 2 + 2 = 4$$

In numeric expressions, multiplication can be represented in different ways. Each expression below is a way to represent "three times five."

3×5 $\quad 3 \cdot 5$ $\quad 3(5)$

In algebraic expressions, the multiplication symbol, ×, could be confused with the variable x. Therefore, writing a number next to a variable is a better way to note multiplication when using variables.

$3x$ means "three times x."

Multiplication of a number and a variable could also be represented using the multiplication dot, $3 \cdot x$, or parentheses, $3(x)$, but $3x$ will be the most common way.

5. Explain why $3 \cdot 5$ cannot be written as 35.
   The dot in the problem represents multiplication. You need to multiply 3 times 5.
6. What is the value of $3x$ if $x = 5$?

$$3 \cdot 5 = 15$$

When substituting values into algebraic expressions, it is helpful to put the substituted value in **parentheses** when the variable is being multiplied by a known value.

7. Find the value of $7m - (3 + b - 12) \div 2$ if $m = 4$ and $b = 15$. Show your work.

$$7(4) - (3 + 15 - 12) \div 2$$
$$28 - (18 - 12) \div 2$$
$$28 - (6) \div 2$$
$$28 - 3 = 25$$

Do you have students who could work as your co-teachers? If so, consider nominating co-teachers to pull their own small groups and have those students lead their small groups through the rest of this activity. Encourage collaboration, questions, and risk-taking as students drive their own learning.

- *How do the parentheses impact the evaluation of the expression?*
- *How would the value change if the parentheses were moved around the first part of the expression?*

11.1.4 Guided Practice: Evaluating Algebraic Expressions

Component Overview: *Students practice evaluating algebraic expressions using the order of operations.*

11.1.4 Guided Practice: Evaluating Algebraic Expressions

1. What is the value of the expression $6 \cdot 2 - (t + 3)$ if $t = 5$?

$$6 \cdot 2 - (5 + 3)$$
$$12 - 8 = 4$$

Engage students in partner work after completing the expressions. Have students evaluate their partners' answers for accuracy and precision. If necessary, engage the class in a discussion to discuss common errors in the work.

2. Find the value of $4 - 10(5 - w) + 4w \cdot 2$ when $w = 3$.

$$4 - 10(5 - 3) + 4(3) \cdot 2$$
$$4 - 10(2) + 12 \cdot 2$$
$$4 - 20 + 24 = 8$$

3. Evaluate $6p + \frac{8}{H} - 7$ for $p = 4$ and $H = 2$.

$$6(4) + \frac{8}{2} - 7$$
$$24 + 4 - 7$$
$$28 - 7 = 21$$

To challenge early finishers, have them evaluate each expression with more variables. Consider substituting variables for decimals or fractions.

11.1.5 Your Turn!

Component Overview: *Students work independently to evaluate algebraic expressions by substituting variables for given numbers.*

11.1.5 Your Turn!

1. Evaluate $13 - 48 \div 2r + 9$ if $r = 8$.

$$13 - 48 \div 2(8) + 9$$
$$13 - 24(8) + 9$$
$$13 - 192 + 9$$
$$-179 + 9 = -170$$

Consider allowing student choice for which problems to solve. This will give students more time to evaluate the expressions and still provides practice for the 11.1.6 Wrap-Up.

2. Find the value of $6 \cdot (q - 3q + 9)$ when $q = 7$.

$$6 \cdot (7 - 3(7) + 9)$$
$$6 \cdot (7 - 21 + 9)$$
$$6 \cdot (-14 + 9)$$
$$6 \cdot (-5) = -30$$

3. What is the value of $7 + \frac{4}{a} - \frac{b}{3} \times 5$ if $a = 1$ and $b = 9$?

$$7 + \frac{4}{1} + \frac{(-9)}{3} \times 5$$
$$7 + 4 + (-3) \times 5$$
$$7 + 4 + (-15) = -4$$

11.1.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their understanding of evaluating algebraic expressions using the order of operations.*

11.1.6 Wrap-Up: Ticket Out the Door

Evaluate $8a + 3b(2a - 4)$ for $a = 2$ and $b = 6$.

$$8(2) + 3(6)(2(2) - 4))$$
$$16 + 18(4 - 4)$$
$$16 + 18(0)$$
$$16 + 0$$
$$16$$

Challenge students who finish early to write an explanation to a sibling or friend on how they used order of operations to evaluate the expression.